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NATIONAL ADVISORY COMMITTEE FOR AERONAUTICS

RESEARCH MEMORANDUM

for the

Air Materiel Command, Army Air Forces

FLIGHT MEASUREMENTS OF FLYING QUALITIES OF A P-47D-30 AIRPLANE (AAF NO. 43-3441) TO DETERMINE LONGITUDINAL STABILITY AND CONTROL AND STALLING CHARACTERISTICS

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SUMMARY

Flight tests have been made to determine the longitudinal stability and control and stalling characteristics of the P-47D-30 airplane. The test results show the airplane to be unstable stick free in any power-on condition even at the most forward center-of-gravity position tested. At the rearward center-of-gravity position tested the airplane also had neutral to negative stick-fixed stability with power on. The characteristics in accelerated flight were acceptable at the forward center-of-gravity position at low and high altitudes except at high speed where the control-force variations with acceleration were high. At the rearward center-of-gravity position, elevator-force reversals were experienced in turns at low speeds, and the force per g was low at all the other speeds. Ample stall warning was afforded in all the conditions tested and the stalling characteristics were very satisfactory except in the approach and wave-off conditions.

INTRODUCTION

At the request of the Air Materiel Command, Army Air Forces, flight tests have been made to determine the flying qualities of a P-47D-30 airplane (AAF No. 43-3441). The results of the tests of longitudinal stability and control and stalling characteristics are presented herein. Data on the lateral and directional stability and control have been presented in a previous report (reference 1).

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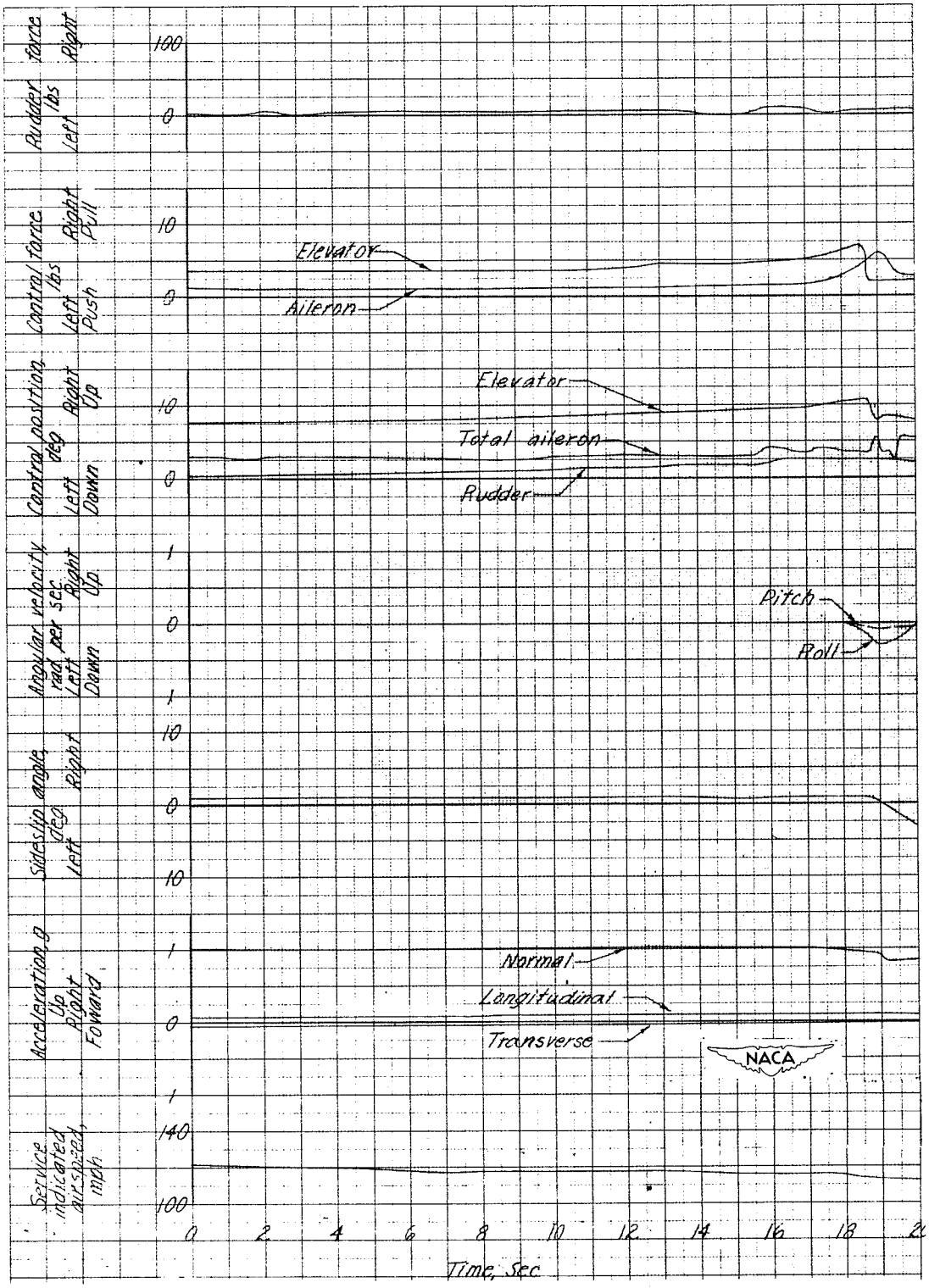


Figure 49.- Time history of a stall approach in the glide condition using elevator alone. Pilot attempted to hold other controls fixed; c.g. at 26.3 percent M.A.C.; gr. wt. 12,400 pounds, 5000 feet altitude.

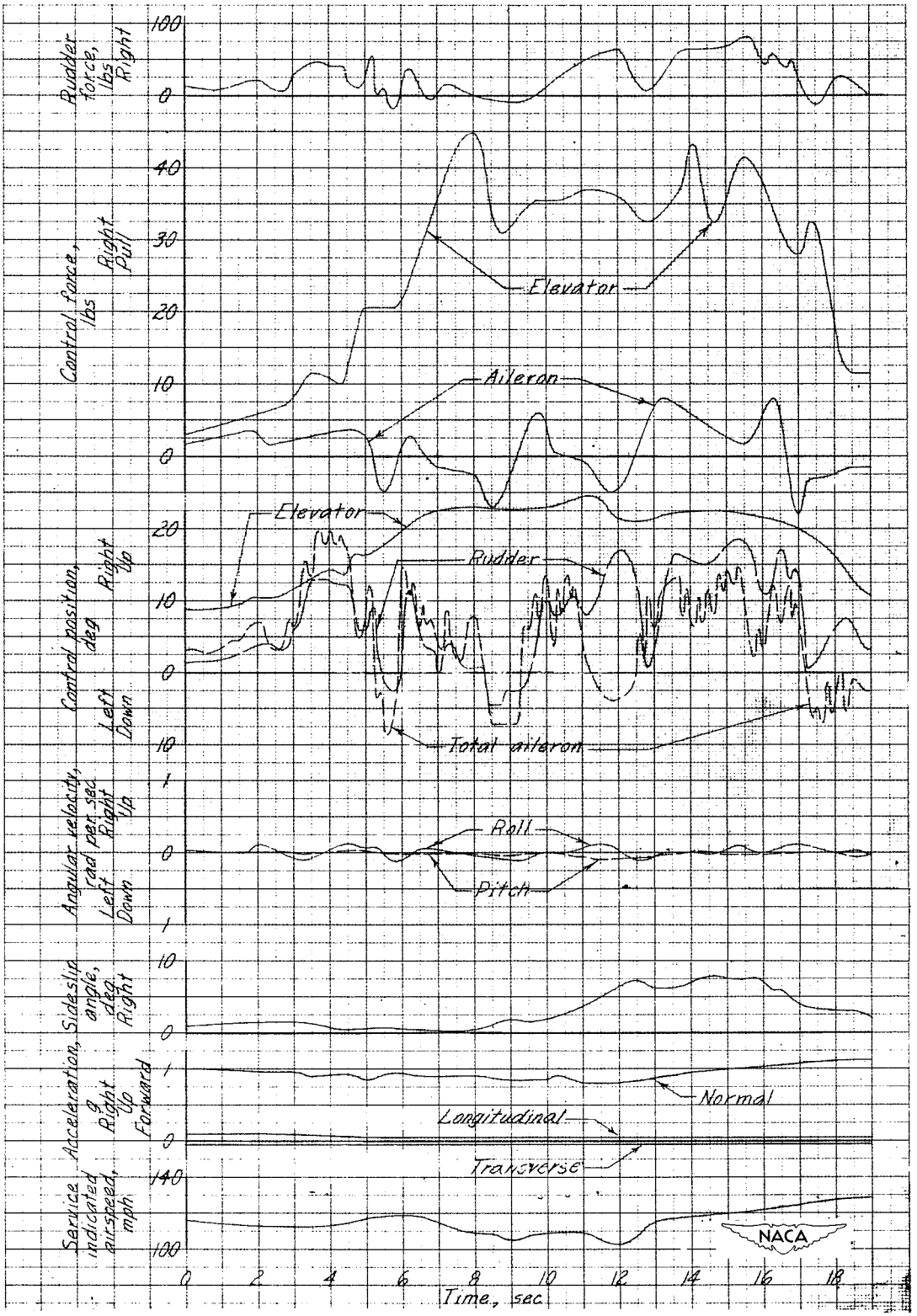
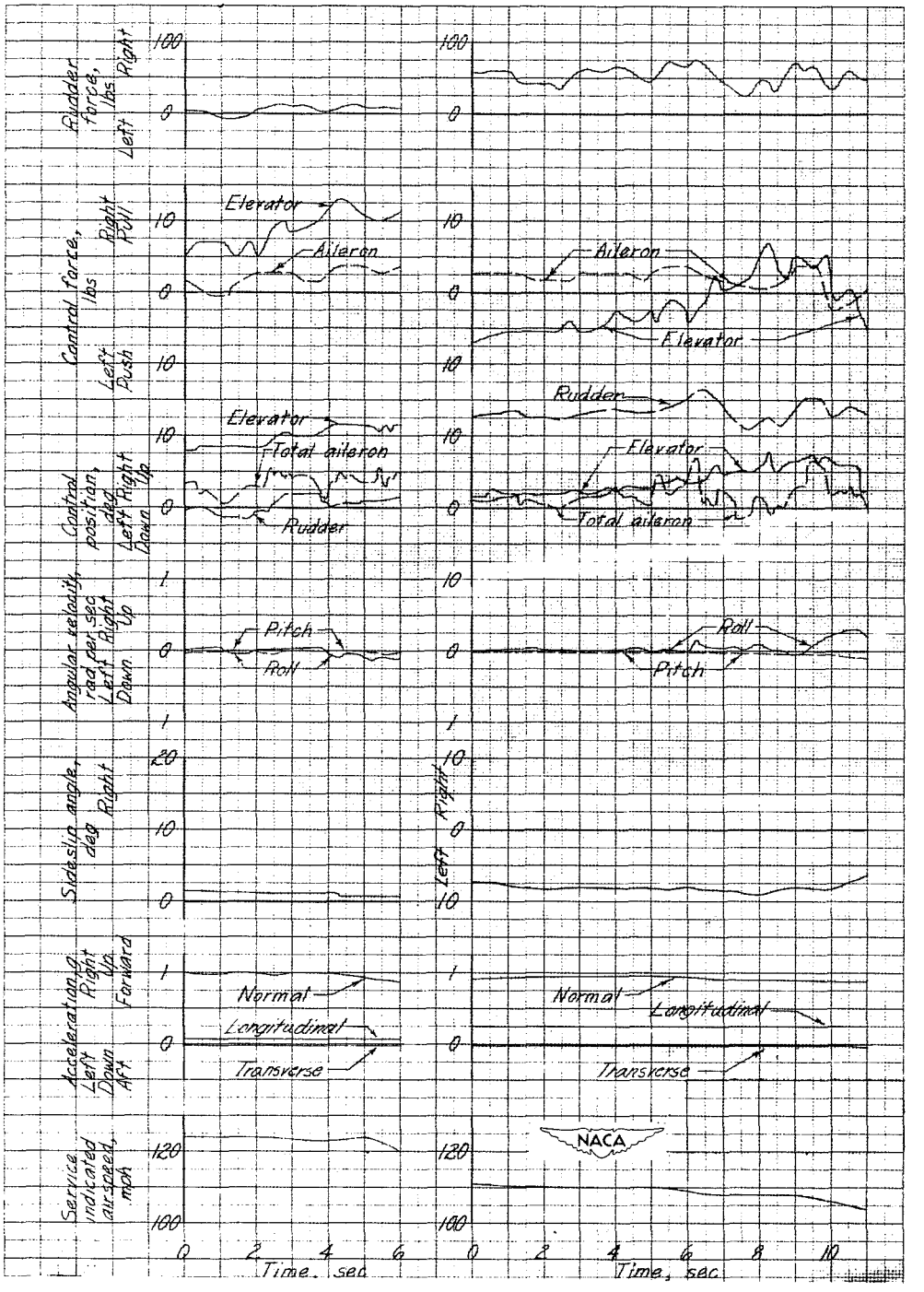


Figure 50.- Time history of a stall approach in the glide condition. All controls used to hold wings level after initial stall; c.g. at 26.3 percent M.A.C.; gr. wt 12,400 pounds; 5000 feet altitude.



(a) Glide condition at 24,000 feet average altitude. Center of gravity 0.264 M.A.C. at 12,770 pounds gross weight.

(b) Power-on clean condition at 26,500 feet average altitude. Center of gravity 0.264 M.A.C. at 12,570 pounds gross weight.

Figure 51.- Time histories of stall approaches, using all controls to hold wings level after initial stall.